

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

COURSE NAME: CSA1613 COURSE CODE: Data Warehousing and Data Mining

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### COURSE OUTCOME COVERED

**CO3:** Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)

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QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

| <b>Criteria</b>   | Understandin g<br>of Concepts<br>(2.5) | Experimental<br>Design (3) | Use of<br>Tools<br>(2) | Analysis of<br>Results (1.5) | Documentation<br>(1) | <b>Total(10<br/>)</b> |
|-------------------|----------------------------------------|----------------------------|------------------------|------------------------------|----------------------|-----------------------|
| <i>Weightag e</i> | 25%                                    | 30%                        | 20%                    | 15%                          | 10%                  | 100%                  |
| <i>Q1</i>         |                                        |                            |                        |                              |                      |                       |
| <i>Q2</i>         |                                        |                            |                        |                              |                      |                       |
| <i>Q3</i>         |                                        |                            |                        |                              |                      |                       |
| <i>Q4</i>         |                                        |                            |                        |                              |                      |                       |
| <i>Q5</i>         |                                        |                            |                        |                              |                      |                       |

**Code of Conduct:**

*I \_G.Mohammad shareef/(192311288)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: G.MOHAMMAD SHAREEF**

**RUBRICS FOR CALCULATION:**

| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                             |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------|
| Understanding of Concepts | 25        | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding unable to integrate concepts |
| Experimental Design       | 30        | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation          |
| Use of Tools              | 20        | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools               |
| Analysis of Results       | 15        | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results           |
| Documentation             | 10        | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                |

**Annexure C****Experiments :**

1.Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

**Photograph:**

| Age of child | A  | B  | C  |
|--------------|----|----|----|
| 5-6 years:   | 18 | 22 | 20 |
| 7-8 years:   | 2  | 28 |    |

9-10 years:                      20                      10                      40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

| Player ID | Player Name | Points Scored |
|-----------|-------------|---------------|
| P1        | Player A    | 12            |
| P2        | Player B    | 15            |
| P3        | Player C    | 14            |
| P4        | Player D    | 10            |
| P5        | Player E    | 18            |
| P6        | Player F    | 20            |
| P7        | Player G    | 22            |
| P8        | Player H    | 13            |
| P9        | Player I    | 11            |
| P10       | Player J    | 35            |

|     |          |    |
|-----|----------|----|
| P11 | Player K | 16 |
| P12 | Player L | 17 |
| P13 | Player M | 19 |
| P14 | Player N | 21 |
| P15 | Player O | 23 |

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.  
(10 Marks)

Consider the Data Set young,high,employed,good,yes young,high,self-employed,average,yes middle,high,employed,good,yes old,medium,employed,good,yes old,low,unemployed,poor,no old,low,self-employed,average,no middle,low,unemployed,poor,no young,medium,employed,average,yes young,low,unemployed,poor,no old,medium,self-employed,good,yes middle,medium,employed,average,yes middle,high,self-employed,good,yes old,medium,unemployed,poor,no young,medium,self-employed,average,yes middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84 Class

B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range. (ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

|             |      |      |      |      |      |      |      |      |      |
|-------------|------|------|------|------|------|------|------|------|------|
| <i>age</i>  | 23   | 23   | 27   | 27   | 39   | 41   | 47   | 49   | 50   |
| <i>%fat</i> | 9.5  | 26.5 | 7.8  | 17.8 | 31.4 | 25.9 | 27.4 | 27.2 | 31.2 |
| <i>age</i>  | 52   | 54   | 54   | 56   | 57   | 58   | 58   | 60   | 61   |
| <i>%fat</i> | 34.6 | 42.5 | 28.8 | 33.4 | 30.2 | 34.1 | 32.9 | 41.2 | 35.7 |

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.

(b) Draw the boxplots for *age* and *%fat*.

(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

## **Answers**

### **1. Relationship Between Age and Photograph Preference :**

#### **Aim:**

To determine whether there is a significant relationship between the age of children and their preference for three photographs using data mining techniques and WEKA.

#### **Dataset:**

| <b>Age Group</b>  | <b>Photograph A</b> | <b>Photograph B</b> | <b>Photograph C</b> |
|-------------------|---------------------|---------------------|---------------------|
| <b>5–6 years</b>  | <b>18</b>           | <b>22</b>           | <b>20</b>           |
| <b>7–8 years</b>  | <b>2</b>            | <b>28</b>           | <b>40</b>           |
| <b>9–10 years</b> | <b>20</b>           | <b>10</b>           | <b>40</b>           |

The given experiment asks whether age and photograph preference are significantly related. It also specifies covariance and correlation calculations for photographs B and C.

#### **Software Used:**

- WEKA
- WEKA Explorer
- Microsoft Word

#### **Procedure in WEKA:**

1. Open WEKA.
2. Select Explorer.
3. Open the prepared ARFF dataset.
4. Select the Preprocess tab.
5. Verify the attributes and number of instances.
6. Go to the Classify tab.

7. Select J48 Decision Tree.
8. Select the photograph preference attribute as the class.
9. Select 10-fold cross-validation.
10. Click Start.
11. Observe the classification results and confusion matrix.

#### Analysis

The distribution of preferences differs among the three age groups. Children in different age groups show different patterns of photograph selection.

Therefore, the data suggests that age and photograph preference may be associated. However, association does not prove that age directly causes a particular preference.

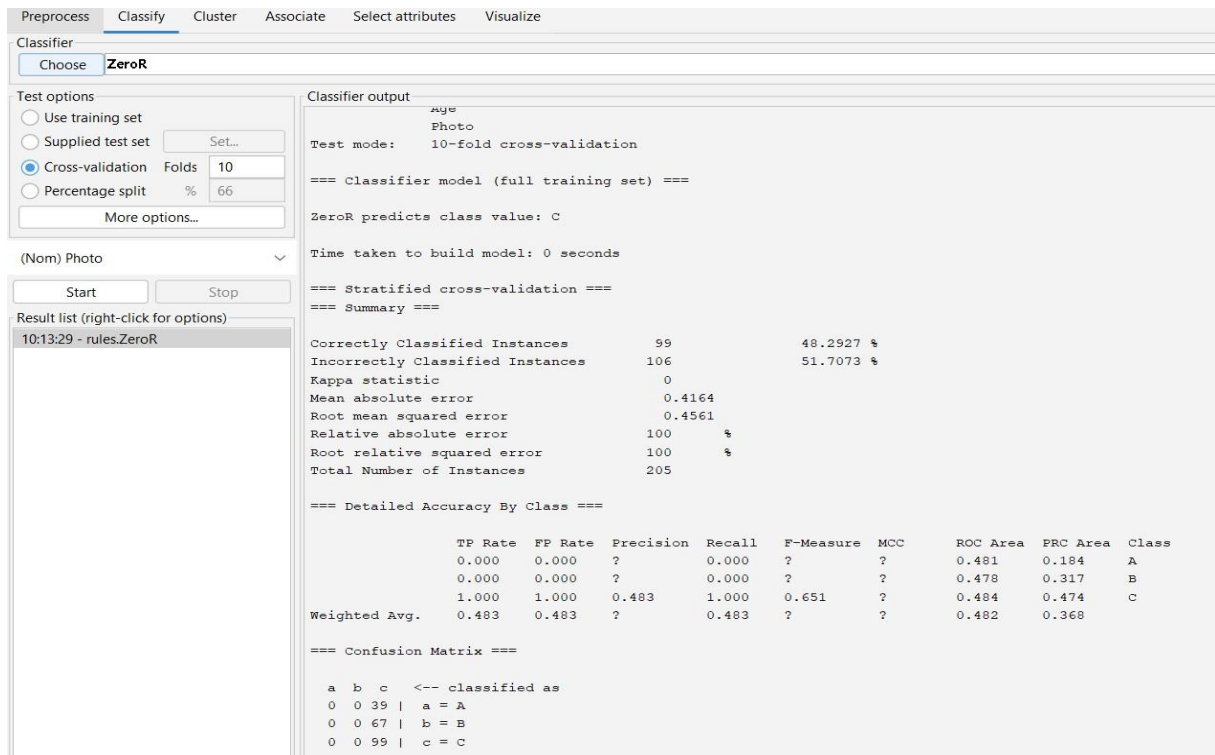
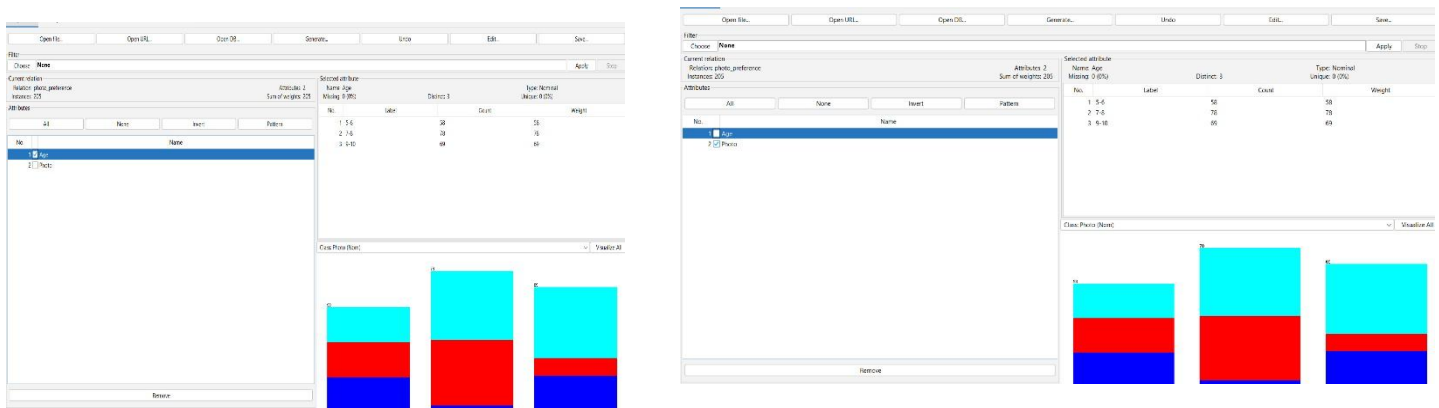
#### **Covariance and Correlation:**

The question specifically requests covariance and correlation between photographs B and C and their corresponding matrices. These calculations are normally performed using statistical functions such as `cov()` and `cor()`; standard WEKA Explorer does not provide those exact R functions directly.

#### **Result:**

The analysis indicates a relationship between age group and photograph preference. The experiment also demonstrates how categorical data can be examined using data mining tools.

## **OUTPUT:**



## 2.Detection of Outliers Using Box Plot:

### Aim:

To visualize the distribution of tennis players' scores and identify possible outliers using a box plot.

### Dataset:

| Player ID | Player Name | Points Scored |
|-----------|-------------|---------------|
| P1        | Player A    | 12            |
| P2        | Player B    | 15            |
| P3        | Player C    | 14            |
| P4        | Player D    | 10            |
| P5        | Player E    | 18            |

|     |          |    |
|-----|----------|----|
| P6  | Player F | 20 |
| P7  | Player G | 22 |
| P8  | Player H | 13 |
| P9  | Player I | 11 |
| P10 | Player J | 35 |
| P11 | Player K | 16 |
| P12 | Player L | 17 |
| P13 | Player M | 19 |
| P14 | Player N | 21 |
| P15 | Player O | 23 |

The above values are provided in the assessment.

#### **Software Used: WEKA Explorer Procedure:**

1. Open **WEKA Explorer**.
2. Select the **Preprocess** tab.
3. Load the tennis dataset.
4. Select the **Points Scored** attribute.
5. Open the visualization option.
6. Generate the distribution/box plot.
7. Observe the median, quartiles and extreme values.
8. Identify values that are considerably separated from the remaining observations.

#### **Observation:**

Most players scored between approximately **10 and 23 points**.

Player J scored **35 points**, which is considerably higher than the other observations.

#### **Result:**

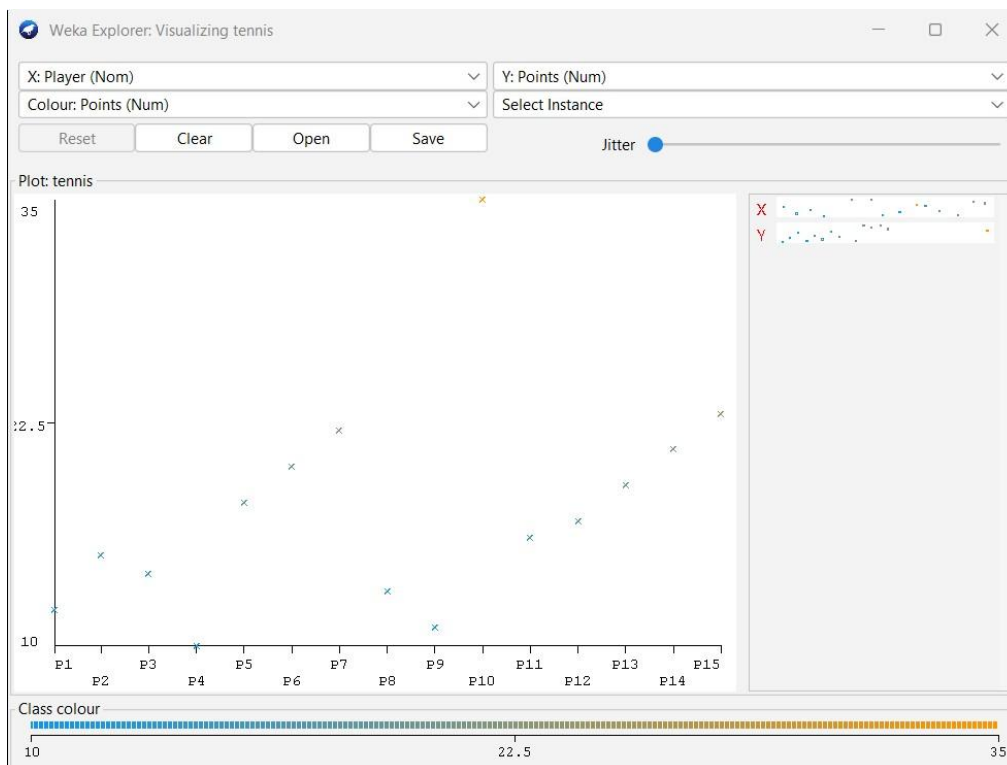
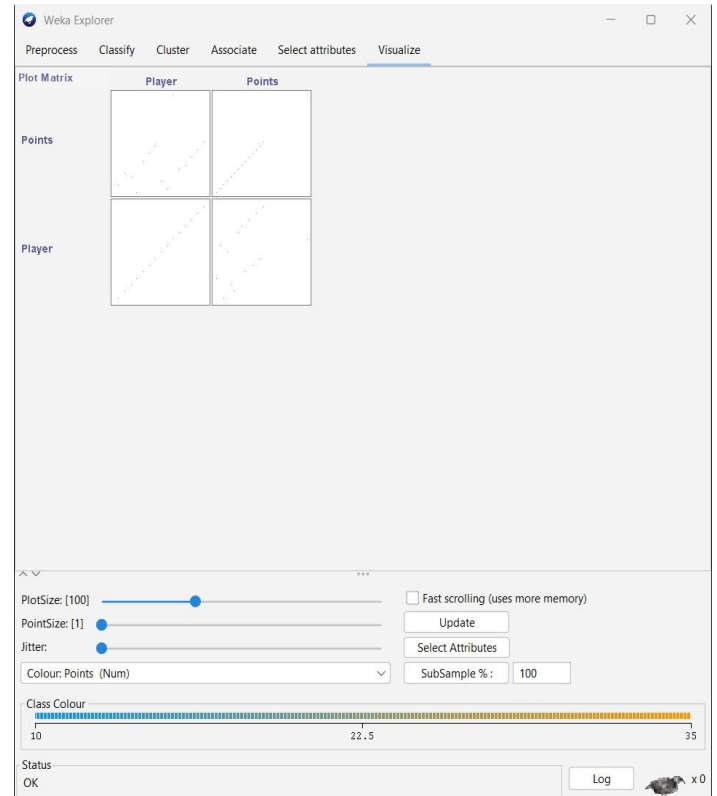
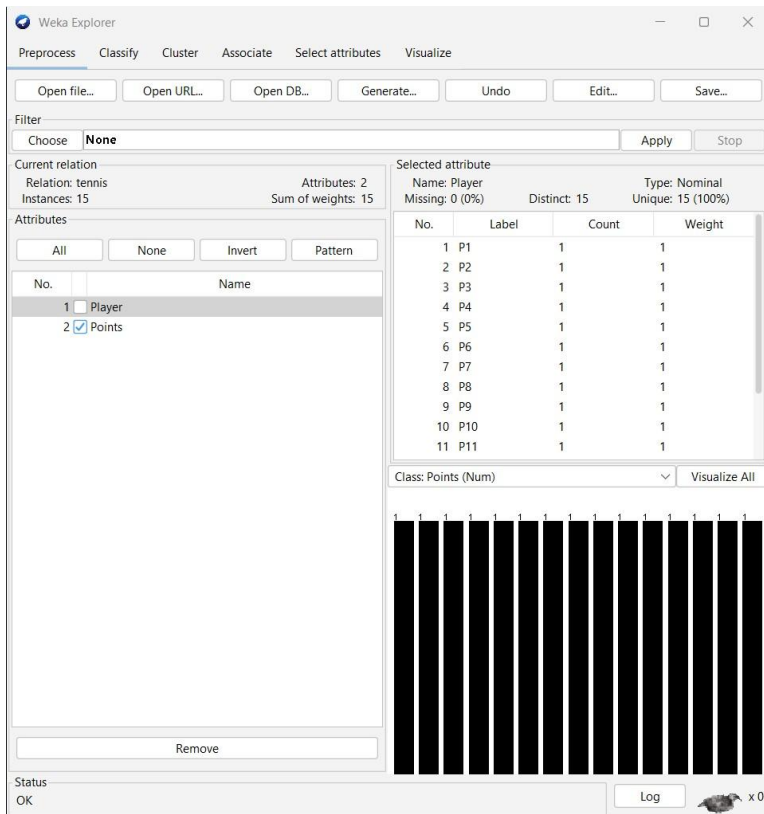
**Player J (P10), with 35 points, is identified as the major high-score outlier.**

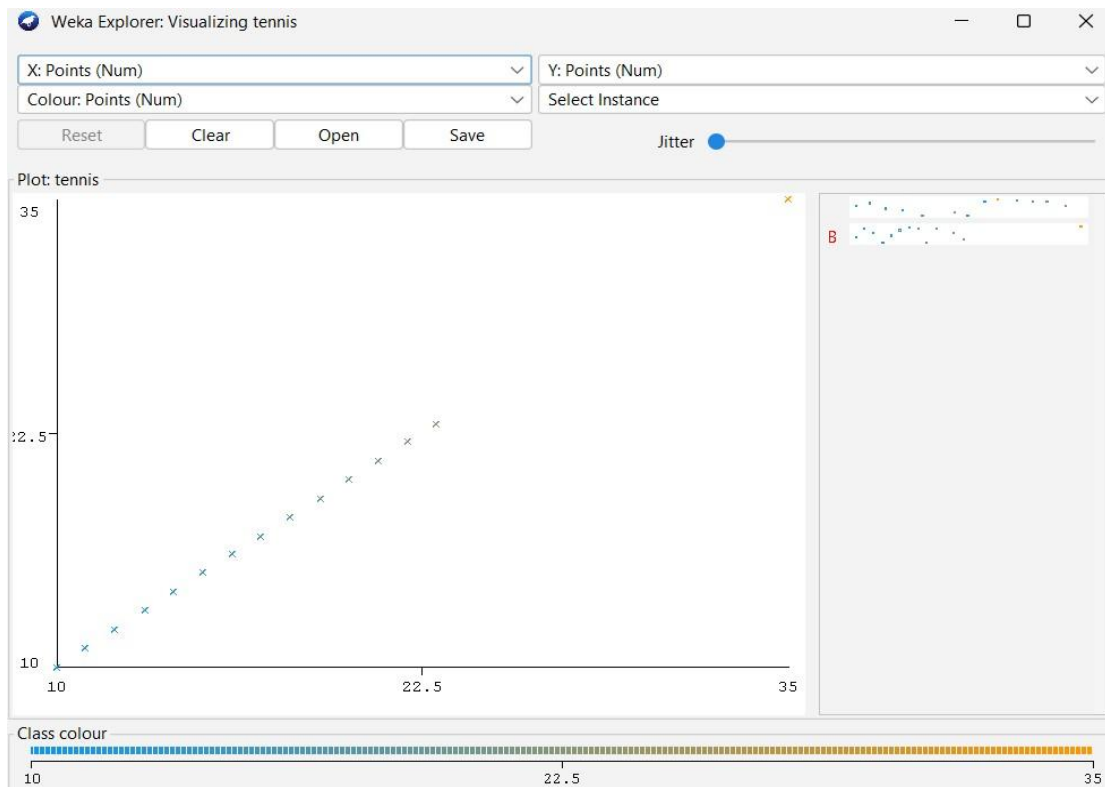
#### **Conclusion:**

The box plot is useful for quickly identifying unusual observations and understanding the distribution of players' scores.

## **OUTPUT:**







### 3. Bank Loan Approval Using Decision Tree

#### Aim:

To build a Decision Tree classifier using WEKA to predict whether a customer's loan should be approved or rejected.

#### Attributes:

The dataset contains:

- Age
- Income
- Employment Status
- Credit Score
- Loan Approval

The supplied dataset contains 15 customer records.

#### Dataset:

| Age   | Income | Employment    | Credit Score | Loan Approval |
|-------|--------|---------------|--------------|---------------|
| young | high   | employed      | good         | yes           |
| young | high   | self-employed | average      | yes           |

|        |        |               |         |     |
|--------|--------|---------------|---------|-----|
| middle | high   | employed      | good    | yes |
| old    | medium | employed      | good    | yes |
| old    | low    | unemployed    | poor    | no  |
| old    | low    | self-employed | average | no  |
| middle | low    | unemployed    | poor    | no  |
| young  | medium | employed      | average | yes |
| young  | low    | unemployed    | poor    | no  |
| old    | medium | self-employed | good    | yes |
| middle | medium | employed      | average | yes |
| middle | high   | self-employed | good    | yes |
| old    | medium | unemployed    | poor    | no  |
| young  | medium | self-employed | average | yes |
| middle | low    | employed      | poor    | no  |

#### **Algorithm Used:**

#### **J48 Decision Tree:**

J48 is WEKA's implementation of the C4.5 decision-tree algorithm.

#### **Procedure in WEKA:**

1. Open **WEKA**.
2. Select **Explorer**.
3. Open the loan.arff dataset.
4. Go to the **Preprocess** tab.
5. Verify that there are 15 instances and 5 attributes.
6. Select the **Classify** tab.
7. Click **Choose**.
8. Select: **trees** → **J48**
9. Set **Loan Approval** as the class attribute.
  - a Select **10-fold cross-validation**.
  - b Click **Start**.
  - c Observe the generated decision tree.
  - d Record the correctly and incorrectly classified instances.

- e Record the accuracy and confusion matrix.

**Expected WEKA Output:** WEKA

produces:

- Decision Tree
- Number of correctly classified instances
- Number of incorrectly classified instances
- Accuracy
- Confusion Matrix
- Detailed accuracy by class

**Analysis:**

The J48 algorithm creates a tree based on the customer attributes and predicts whether the loan should be approved.

Customers with favorable combinations of income, employment status and credit score are generally classified as **Yes**, while customers with low income, poor credit score and unfavorable employment status are more likely to be classified as **No**.

**Result:**

A Decision Tree classifier was successfully developed using the **J48 algorithm in WEKA** to predict loan approval.

**WEKA Screenshot 1:** Preprocess tab

**WEKA Screenshot 2:** J48 classification output **WEKA**

**Screenshot 3:** Generated decision tree

**Conclusion:**

WEKA's J48 algorithm can be effectively used for classification problems such as bank loan approval.

**Output:**



## 4.Comparison of Mathematics Class Performance:

### Aim:

To compare two Year 9 classes using mean, median and range and visualize their distributions using a box plot.

### Dataset:

#### Class A:

76, 35, 47, 64, 95, 66, 89, 36, 84

#### Class B:

51, 56, 84, 60, 59, 70, 63, 66, 50

### Software Used: WEKA Explorer

### Procedure:

1. Create a CSV file containing the class and score.
2. Open **WEKA Explorer**.
3. Select **Preprocess**.
4. Open the CSV dataset.
5. Verify the Class and Score attributes.
6. Use the visualization facility to examine the score distribution.
7. Compare the distributions of Class A and Class B.

### Calculated Results:

| Measure | Class A | Class B |
|---------|---------|---------|
| Mean    | 65.78   | 62.11   |
| Median  | 66      | 60      |
| Range   | 60      | 34      |

### Analysis:

- Class A has the **higher mean**.
- Class A has the **higher median**.
- Class A has the **larger range**.
- Class A therefore performed better on average.
- The larger range indicates greater variation in Class A scores.

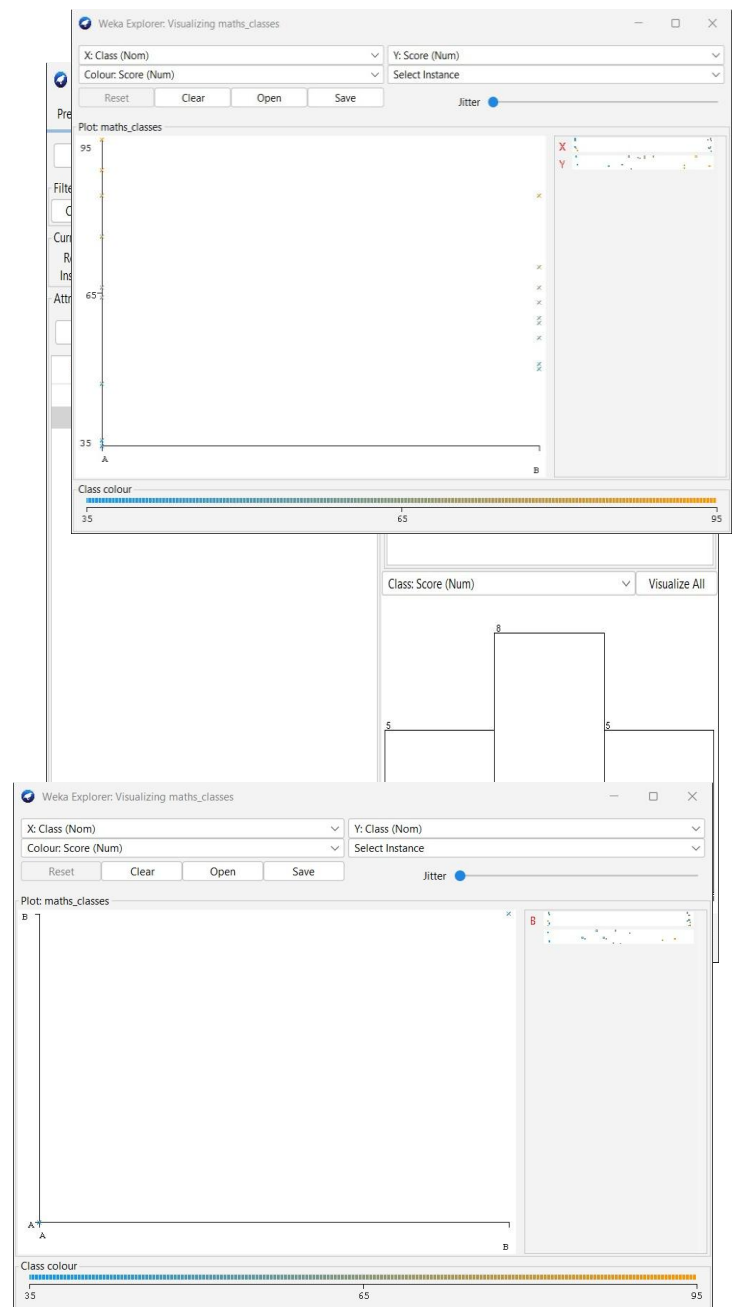
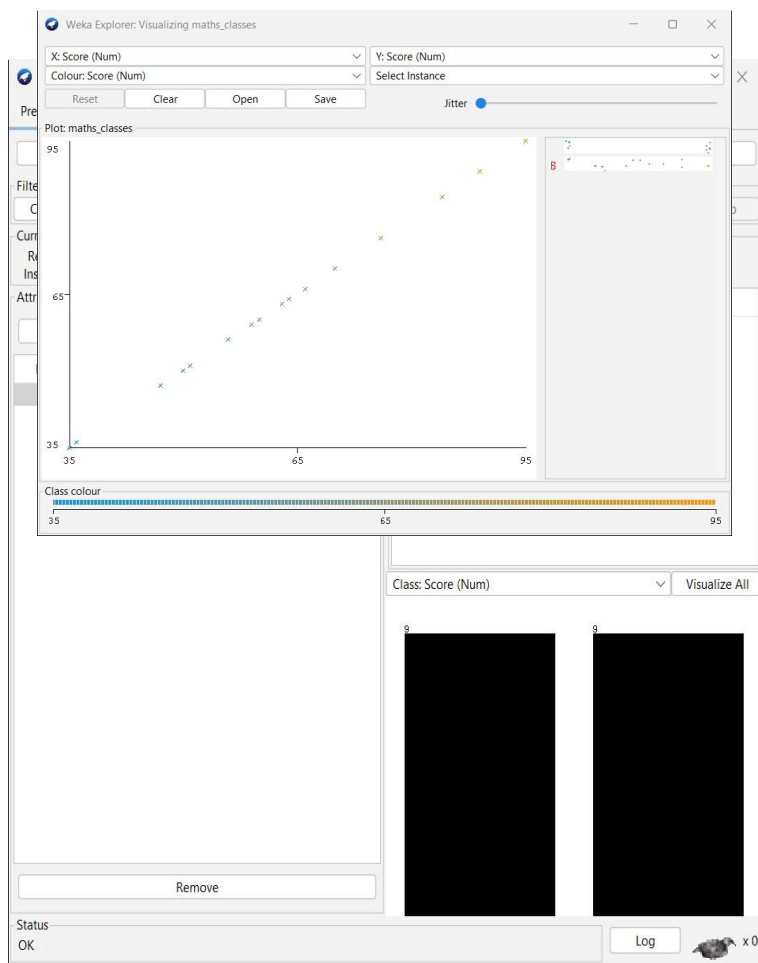
### Result:

**Class A scored higher in mean and median.**

### Conclusion:

Class A shows better overall performance, while Class B has a smaller spread of scores.

### Output:



## 5. Analysis of Age and Body Fat

### Aim:

To calculate and visualize the statistical characteristics of age and body-fat data using WEKA.

### Dataset:

The question specifies data for 18 randomly selected adults, with age and body-fat measurements. The supplied PDF text does not contain the actual 18

numerical observations, so the values should be entered exactly as provided in the original question/table. Software

### **Used: WEKA Explorer**

#### **Attributes:**

- Age
- Body Fat (%) **Procedure:**

#### **Step 1: Create Dataset**

Create a CSV file in the following format:

Age,BodyFat

<age1>,<fat1>

<age2>,<fat2>

...

<age18>,<fat18> Save it

as: **hospital.csv**

#### **Step 2: Load Dataset:**

1. Open **WEKA Explorer**.
2. Select **Preprocess**.
3. Click **Open file**.
4. Select hospital.csv.
5. Verify the two numeric attributes.

#### **Step 3: Statistical Analysis:**

Select the **Age** attribute and record:

- Mean
- Standard deviation
- Minimum
- Maximum

Repeat the process for **Body Fat**.

#### **Step 4: Box Plot:**

Use the visualization facilities to examine the distribution of:

- Age
- Body Fat

The box plot helps identify:

- Median
- Quartiles
- Spread
- Possible outliers

#### **Step 5: Scatter Plot:**

Select:

**X-axis → Age**



### **Y-axis → Body Fat**

Observe the distribution of points.

### **Scatter Plot Interpretation:**

If the points generally increase from left to right, there is a positive relationship between age and body fat.

If the points generally decrease, there is a negative relationship. If the points are widely scattered, there may be little or no clear linear relationship.

The exact conclusion should be based on the actual 18 observations.

### **Q-Q Plot:**

The question also asks for a Q-Q plot. Standard WEKA Explorer does not directly provide the requested Q-Q plot in the same way as dedicated statistical software, so the plot should not be claimed as a native WEKA output unless the installed WEKA version/add-on provides it.

### **Result:**

The age and body-fat data can be statistically summarized and visualized using WEKA. Box plots identify distribution and possible outliers, while the scatter plot helps examine the relationship between age and body fat.

### **Conclusion:**

WEKA provides useful facilities for exploratory data analysis, including attribute statistics and visualization.

## **OVERALL CONCLUSION:**

The experiments demonstrate the application of data mining and statistical analysis techniques using **WEKA**. The experiments include categorical-data analysis, outlier detection, Decision Tree classification, performance comparison and visualization. The **J48 Decision Tree** experiment demonstrates how WEKA can be used to construct a predictive classification model. Box plots and scatter plots provide useful visual methods for understanding data distributions and relationships.

## **SOFTWARE USED:**

- WEKA
- WEKA Explorer
- Microsoft Word

## **GENERAL WEKA WORKFLOW:**

The common procedure followed in these experiments is:

**WEKA → Explorer → Preprocess → Load Dataset → Select Attribute → Visualize / Classify → Analyze Results** For classification:  
**WEKA → Explorer → Preprocess → Classify → Choose J48 → Select Class → 10-Fold Cross-Validation → Start**

## FINAL OBSERVATIONS:

| Experiment | Technique                 | Main Result                                               |
|------------|---------------------------|-----------------------------------------------------------|
| 1          | Age vs Photograph         | Age group and preference                                  |
|            | Preference                | show different patterns<br>Player J with 35 points is an  |
| 2          | Box Plot                  | obvious high-score outlier                                |
| 3          | J48 Decision Tree         | Loan approval can be predicted from customer attributes   |
| 4          | Mean, Median, Range       | Class A has higher mean and median                        |
| 5          | Statistical Visualization | Age and body-fat distributions can be analyzed using WEKA |

## OUTPUT:

